Bud seed cues comparison model methods outline

Mao

August 11, 2026

1 Methods

1. We used different data and models for budburst and seed germination.
   (a) We have complete data for 191 species for the budburst model.
   (b) We have complete data for 329 angiosperm species and 107 gymnosperm species for the seed germination model.

2. We modeled budburst as a function of forcing, chilling, and photoperiod, accounting for phylogeny:

$$\hat{y}_i = \alpha_{sp[i]} + \beta_{force,sp[i]} \cdot forcing_i + \beta_{chill,sp[i]} \cdot chilling_i + \beta_{photo,sp[i]} \cdot photoperiod_i$$

   where $sp[i]$ is the species of observation i, and species-level intercepts and slopes (α_{sp}, $\beta_{\text{force},sp}$, $\beta_{\text{chill},sp}$, $\beta_{\text{photo},sp}$) are modeled according to species phylogenetic relationships.

3. We ran the model with forcing and chilling for seed germination considering phylogeny in the model

4. We modeled seed germination as a function of forcing and chilling, accounting for phylogeny, following the same structure as above (excluding the photoperiod term). The response variable was standardized to the $[0, 1]$: for germination measured as a percentage, we used the proportion directly; for germination recorded as a binary outcome, we coded observations as germinated (1) or not (0).

5. After fitting both models, we extracted the species-level parameter estimates and compared the chilling and forcing effects on budburst versus seed germination for the 38 species overlapping between the two datasets.